

SUWON AB, Korea, Republic of

WMO: 471200

Lat: 37.239N

Lon: 127.007E

Elev: 88

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	10.7	-1.6	5.1	12.5	2.8	6.4	14.7	23.7	26.3	20.5	25.4	1.2	300	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.5	91.6	78.8	89.4	77.3	86.4	75.6	80.1	88.3	79.1	86.5	77.9	84.3	6.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.8	84.9	77.0	141.3	83.2	75.6	134.4	81.3	43.9	89.0	42.7	86.4	41.4	83.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	15.7	13.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			4.3	82.4	5.5	1.5	0.4	83.5	-2.8	84.4	-5.9	85.2	-9.9	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	25.5	30.7	41.8	53.7	63.9	72.2	78.2	79.4	69.7	58.0	45.3	33.3	
	DBStd	19.10	7.19	8.14	5.86	6.79	4.68	4.07	4.06	4.18	5.28	6.69	7.37	7.36	
	HDD50	2301	758	541	257	36	0	0	0	0	0	18	174	517	
	HDD65	5142	1223	961	718	343	77	4	0	0	16	228	590	982	
	CDD50	3929	0	0	4	148	432	667	875	912	592	265	34	0	
	CDD65	1293	0	0	0	5	43	221	410	447	158	9	0	0	
	CDH74	10121	0	0	0	53	335	1394	3312	3950	999	78	0	0	
	CDH80	3224	0	0	0	4	42	363	1108	1494	210	3	0	0	
Wind	WSAvg	4.2	4.2	4.8	5.2	5.2	4.5	3.9	4.1	3.7	3.3	3.6	3.8	3.9	
Precipitation	PrecAvg	51.0	1.0	1.2	1.9	3.6	3.4	4.7	13.2	11.0	5.8	2.2	2.1	0.9	
	PrecMax	80.5	3.5	4.6	5.0	15.6	9.6	16.3	22.4	33.5	23.0	10.4	5.9	2.1	
	PrecMin	33.5	0.1	0.0	0.2	0.3	0.5	0.2	5.7	2.1	0.1	0.0	0.4	0.0	
	PrecStd	10.7	0.9	1.0	1.1	3.4	2.1	4.1	5.5	6.1	4.7	2.4	1.4	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	55.3	64.7	78.9	84.2	89.7	95.0	95.0	88.2	79.1	67.9	56.0	
		MCWB	39.3	44.9	51.3	61.8	66.6	72.1	79.7	80.0	74.8	65.3	58.5	48.2	
	2%	DB	42.7	50.3	60.4	75.0	80.7	86.3	91.4	92.8	85.6	75.6	64.1	51.5	
		MCWB	37.5	43.4	48.2	61.0	64.9	70.9	79.1	79.6	73.4	63.7	55.1	44.7	
	5%	DB	39.6	46.7	56.9	71.2	77.8	84.2	88.2	90.0	82.4	73.2	60.8	48.1	
		MCWB	35.5	41.4	47.0	57.4	63.3	69.9	77.4	78.4	71.4	62.1	52.7	43.5	
	10%	DB	37.3	43.2	53.3	67.0	75.1	81.8	86.0	87.8	79.3	69.7	57.3	44.8	
		MCWB	33.8	38.1	44.8	55.3	61.8	69.2	76.4	77.6	69.6	60.4	51.2	40.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	48.5	53.8	66.8	71.1	76.2	81.7	82.2	77.6	68.1	60.2	49.8
			MCDB	44.1	52.3	60.3	73.4	78.4	84.3	90.3	91.5	83.7	74.6	65.4	53.3
2%		WB	38.4	44.3	50.0	62.2	67.9	73.8	80.0	80.6	75.7	66.0	57.2	46.1	
		MCDB	41.3	48.1	58.1	71.5	76.0	81.2	88.1	89.1	81.5	72.8	62.1	49.2	
5%		WB	36.2	41.4	47.5	59.4	65.8	72.4	79.0	79.6	73.9	63.7	54.5	43.5	
		MCDB	39.5	47.0	54.5	68.2	74.0	79.8	86.0	87.5	79.5	70.5	58.8	47.3	
10%		WB	33.7	38.6	45.2	56.6	63.9	70.9	77.8	78.6	71.9	61.6	51.9	40.8	
		MCDB	36.3	42.9	52.0	65.0	71.6	78.2	83.9	85.6	78.0	67.8	56.5	44.6	
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.8	19.3	22.3	19.9	17.2	13.1	14.5	18.1	20.9	19.3	17.3	
		MCDBR	18.0	20.5	25.1	27.3	24.5	21.6	17.1	17.8	19.0	24.1	22.6	20.4	
		MCWBR	14.1	14.1	15.3	14.8	11.5	9.5	7.5	7.6	9.0	13.3	14.9	15.0	
	5% WB	MCDBR	17.4	18.4	22.6	22.7	19.7	17.3	14.4	16.1	15.4	20.8	19.1	18.6	
		MCWBR	14.2	13.0	15.2	13.7	11.1	8.8	7.0	7.4	8.7	13.0	14.4	15.0	
Clear-Sky Solar Irradiance	taub	0.440	0.490	0.573	0.613	0.626	0.698	0.662	0.582	0.510	0.499	0.482	0.437		
	taud	1.947	1.858	1.714	1.675	1.696	1.618	1.740	1.928	2.053	1.958	1.944	1.973		
	Ebn at Noon	220	226	220	221	221	206	212	226	236	221	205	211		
	Edh at Noon	44	55	69	76	76	82	72	59	49	49	44	41		
All-Sky Solar Radiation	RadAvg	788	1034	1358	1580	1774	1649	1307	1384	1307	1139	778	692		
	RadStd	48	84	136	115	164	165	196	198	142	83	79	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+90

Nomenclature: See separate page